

EXAM 3

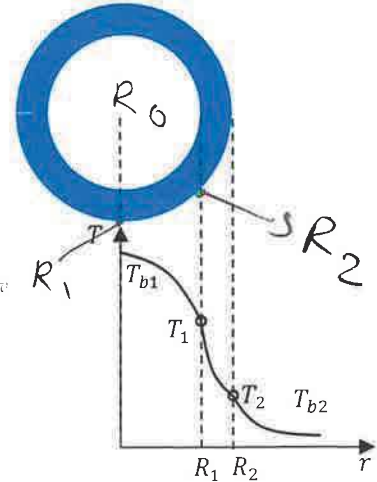
CLOSED BOOK, TIME: 120 MINUTES

1. Consider a cylindrical pipe with an inner radius R_1 , an outer radius R_2 , and a constant thermal conductivity k .

- 1.1. Using the equation of energy in cylindrical coordinates, derive an equation for the steady-state temperature profile and the heat flux on the inner surface of the pipe in terms of T_1 and T_2 , the temperatures on the inner and outer surface of the pipe.

$$\rho \hat{C}_p \left(\frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + \frac{v_\theta}{r} \frac{\partial T}{\partial \theta} + v_z \frac{\partial T}{\partial z} \right) = k \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \Phi_v$$

- 1.2. A fluid with bulk temperature T_{b1} and heat transfer coefficient h_1 is in contact with the inner surface of the pipe while a second fluid with bulk temperature T_{b2} and heat transfer coefficient h_2 is in contact with the outer surface of the pipe. Derive an equation for the overall heat transfer coefficient U_1 for heat exchange between the two fluids with reference to the inner radius R_1 .



Hint: Heat flux on the inner surface is $q_r|_{r=R_1} = h_1(T_{b1} - T_1) = U_1(T_{b1} - T_{b2})$ and the one on the outer surface is $q_r|_{r=R_2} = h_2(T_2 - T_{b2}) = \frac{R_1}{R_2} q_r|_{r=R_1}$.

$$0 = k \left[\frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) \right]$$

$$0 = k \frac{d^2 T}{dr^2}$$

$$-q_o = k \frac{dT}{dr}$$

$$\int_{r_1}^{r_2} dr = \int_{T_1}^{T_2} \frac{k}{-q_o} dT$$

$$r_2 - r_1 = \frac{k}{-q_o} (T_2 - T_1)$$

$$r_2 - r_1 = \frac{k}{q_o} (T_1 - T_2)$$

$$q_{r1} = -k \frac{dT}{dr}$$

By Newton's law of cooling.

$$\frac{q_o}{h_1} = (T_{b1} - T_1)$$

$$\frac{q_o}{h_2} = (T_2 - T_{b2})$$

Q1 Name: Parth Patel.

$$T_{b1} - T_1 + T_2 - T_{b2} + T_1 - T_2 = q_o \left(\frac{1}{h_1} + \frac{1}{h_2} + \frac{(x_2 - x_1)}{K} \right)$$

(1.2) $T_{b1} - T_{b2} = q_o \left(\frac{1}{h_1} + \frac{1}{h_2} + \frac{(x_2 - x_1)}{K} \right)$

given $q_{x|_{x=R_1}} = U_1 (T_{b1} - T_{b2})$

$$\frac{q_{x|_{x=R_1}}}{U_1} = q_o \left(\frac{1}{h_1} + \frac{1}{h_2} + \frac{(x_2 - x_1)}{K} \right)$$

given $q_{x|_{x=R_1}} = h_1 (T_{b1} - T_1)$

$$\frac{h_1 (T_{b1} - T_1)}{U_1} = q_o \left(\frac{1}{h_1} + \frac{1}{h_2} + \frac{(x_2 - x_1)}{K} \right)$$

$$U_1 = \frac{h_1 (T_{b1} - T_1)}{q_o \left(\frac{1}{h_1} + \frac{1}{h_2} + \frac{(x_2 - x_1)}{K} \right)}$$

Name:

Parthiv Patel

2.3

$$\frac{Q(\eta, \tau)}{\cos\left[\frac{2n+1}{2}, \pi\eta\right]} = \sum_{n=0}^{\infty} f_n(\tau)$$

$$C_{A0} = \frac{(C_2 + C_1(\eta b))t}{-2b^2(1-Q)} = Q(\eta, \tau) = 1 - \frac{(C_2 + C_1(\eta b))t}{-tb^2C_{A0}}$$

$$\left[1 - \frac{(C_2 + C_1(\eta b))t}{-tb^2C_{A0}}\right] \div \cos\left[\frac{2n+1}{2}, \pi\eta\right] = \sum_{n=0}^{\infty} f_n(\tau)$$

2.4

$$Q = aN_A \Big|_{y=b} = -a \overset{\text{surface area}}{D_{AB} \frac{dC_A}{dy}} \Big|_{y=b}$$

Name: